

MOTIVATION

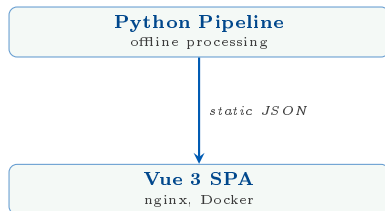
Despite Informatics Europe (IE) maintaining rich time-series data on enrollment, graduation, and female representation, that data lives in Excel workbooks requiring manual effort to interpret. Existing portals offer only tabular views; geographic and temporal patterns remain hidden.

Goal: a unified browser-based platform transforming raw spreadsheets into interactive, multi-level visualizations for researchers, administrators, and policy makers.

DATA SOURCES

- **Members-DATA.xlsx** — per-institution BSc/MSc/PhD enrollment, degrees, female counts (IE members)
- **ETER export** — EU-wide data across ≈3500 institutions, 41 countries
- **IE Website scraper** — live member list + geocoords via OSM Nominatim
- **Wikidata SPARQL** — population figures for per-capita normalization
- **YAML files** — manually curated dept-level data (CZ, DE, NO, PT, LV)

SYSTEM ARCHITECTURE



No backend at runtime — the SPA fetches pre-built JSON files. Deployment via a single Docker Compose.

INTERACTIVE MAP



Fig. 1: Europe-wide view. Markers cluster automatically; green = IE member, blue = other.

CHOROPLETH LAYER

Nine configurable combinations: {BSc, MSc, PhD} × {enrolled, first-year, degrees} with four metrics — *total*, *female count*, *female share (%)*, *per-million population*.

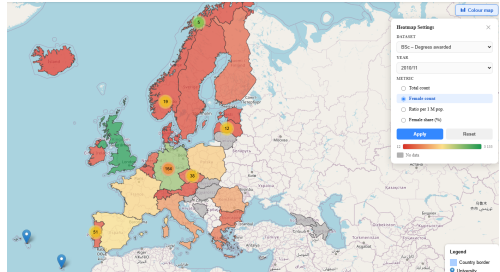


Fig. 3: Female BSc count, 2010/11.

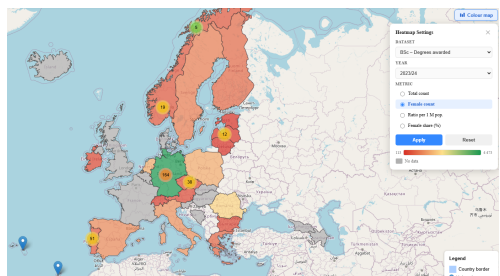


Fig. 4: Same metric, 2023/24. Absolute counts rose; relative shares barely moved.

STATISTICAL CHARTS

Enrollment Trends — stacked bars separating first-year vs. returning cohorts for bachelor's level.

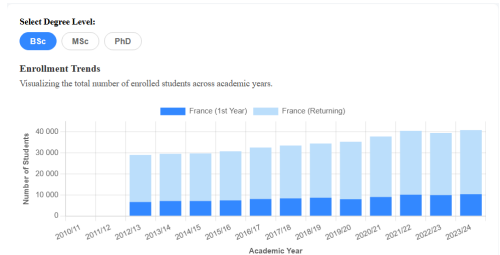


Fig. 5: BSc enrollment, France. Dark = first-year; light = returning.

Female Representation — dual line chart (BSc & MSc) exposing persistent gender disparities at programme level.

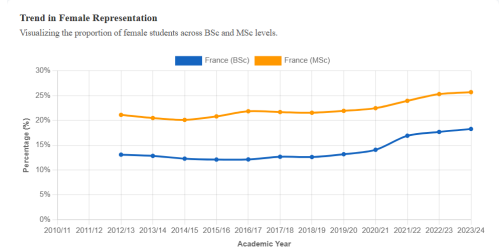


Fig. 6: France BSc (blue) vs. MSc (orange) female share, 14 years. Both below 20%.

EUROPEAN CONTEXT & DENSITY

Three views: **Doughnut** (country shares), **Absolute Scale** (raw), **Density** (per million — small Nordic nations rank highest).

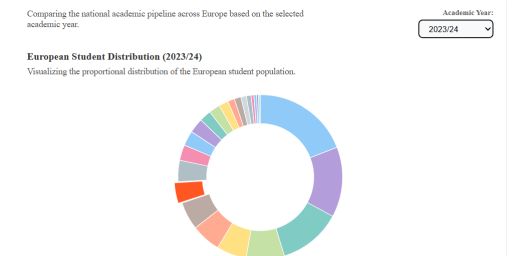


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SIDE-BY-SIDE COMPARISON

Any two countries or institutions can be placed in a **Compare** modal using *colour hue* for degree levels and *line style* (solid vs. dashed) to distinguish entities. Covers enrollment, degrees, and female share at BSc & MSc level.

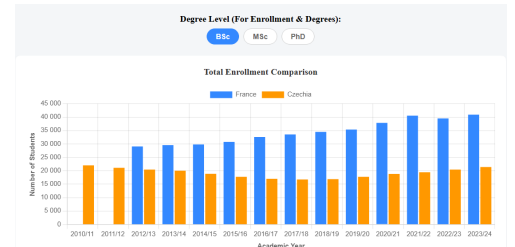


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KEY FINDINGS

- Most European countries show <25% **female share** at BSc level — nearly unchanged over 14 years
- Absolute female counts *grew*, masking the stagnant relative share
- **Per-capita normalization shifts the picture**: small Nordic/Baltic nations rank highest once population size is accounted for

LIMITATIONS & FUTURE WORK

- Manual pipeline re-run on each update; ETER may lag 2+ years
- Members-DATA = IE members only; fuzzy matching risk

Future improvements: CI/CD auto-refresh, choropleth animation, institution-type filter, mobile layout, CSV/PNG/PDF export.